Signal-Adaptive Retrieval for Domain QA: Design, Deployment, and Evaluation — Traditional Chinese Bible QA as an Example

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Abstract

Domain-specific question answering with large language models (LLMs) incurs high API costs and raises data-privacy concerns. We design, deploy, and evaluate a signal-adaptive RAG architecture for Traditional Chinese Bible QA, integrating PostgreSQL, Qdrant, and Neo4j with a 6-route retrieval system driven by query signal features. On a 100-question benchmark spanning five question types, an ablation against a semantic-only baseline shows the signal-adaptive retrieval is the dominant quality lever, raising hit rate from 0.81 to 0.96 and Answer Coverage from 0.618 to 0.75. Holding retrieval constant, a locally-deployed Gemma 4 E4B matches Claude Haiku 4.5—6 of 7 metrics differs by under 4 percentage points, with neither model dominating. These results show retrieval design, not model scale, is the primary determinant of domain QA quality, enabling cost-free, privacy-preserving deployment.

I. INTRODUCTION

Large language models (LLMs) excel at general question answering, but domain-specific deployment faces two core challenges: (1) high commercial API costs that create barriers for resource-constrained organizations, and (2) limited coverage of specialized domain knowledge during pre-training, resulting in factual gaps that persist across model sizes.

Recent studies show that even state-of-the-art LLMs score poorly on faith-related dimensions [1], highlighting legitimate concerns within the Christian community regarding whether LLMs can provide answers grounded in Scripture. Therefore, we investigate these challenges in the context of Traditional Chinese biblical question answering, a domain where factual accuracy and theological faithfulness are paramount.

Retrieval-Augmented Generation (RAG) [2] mitigates factual errors by supplementing generation with retrieved evidence. While existing RAG frameworks may combine knowledge graphs [3], dense retrieval [4], and lexical methods [5] across different architectures [6], most benchmarking varies retrieval under only a single model. Our study benchmarks against different architectures and model sizes. The contributions are threefold:

1) A signal-adaptive 6-route RAG architecture integrating PostgreSQL, Qdrant, and Neo4j, using a decision-tree router to select optimal engine combinations based on six query features.

2) A controlled ablation showing that signal-adaptive retrieval improves hit rate by 15 percentage points (pp) and Answer Coverage by 10 pp over a semantic-only baseline.

3) Empirical evidence showing that, with retrieval held constant, the local, zero-cost Gemma 4 E4B [7] matches Claude Haiku 4.5 [8], as evaluated via the RAGAS framework [9] spanning 100 curated biblical domain questions across 5 categories.

II. SYSTEM ARCHITECTURE

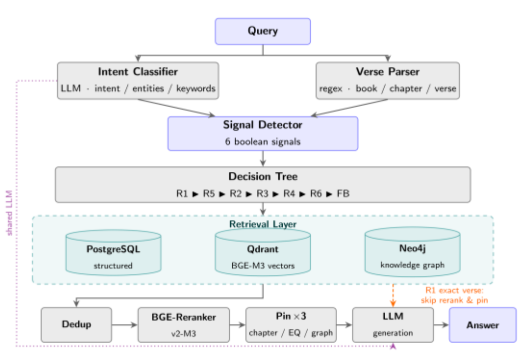

Fig. 1. Signal-adaptive 6-route RAG pipeline.

A. Data Processing Pipeline

The Traditional Chinese Bible's 66 books undergo hierarchical chunking—Book (66) → Chapter (1,189) → Pericope (2,779) → Chunk (431), total 4,465 corpus nodes, with token-aware splitting (512–768 tokens) on verse boundaries. CKIP Transformers [10] perform Chinese NER with LLM validation, yielding 9,122 entities across 6 types (Person 2,418, Object 2,200, Event 1,712, Place 1,299, Theme 987, Group 506) linked by 41,140 MENTIONS edges. BGE-M3 [11] embeds 34,072 passages (1024-dim) at pericope, chunk, and verse granularity.

B. Triple-Database Architecture

PostgreSQL stores the structured corpus hierarchy—66 books, 1,189 chapters, 2,779 pericopes, 431 chunks—for exact lookup. Qdrant provides dense vector search over 34,072 embedding points. Following Graph RAG principles [12], Neo4j maintains a heterogeneous knowledge graph of 13,587 nodes and 56,453 relationships organized into two layers. The node set comprises 4,465 corpus nodes mirroring the PostgreSQL hierarchy (66 books, 1,189 chapters, 2,779 pericopes, 431 chunks) and 9,122 entity nodes from NER. Relationships fall into three categories: 8,355 corpus structural edges (CONTAINS, NEXT and NEXT_BOOK, CROSS_REFERENCES), 41,140 MENTIONS edges that anchor entities to the corpus nodes where they appear, and 6,958 entity-to-entity fact edges supporting multi-hop reasoning.

C. Signal-Adaptive 6-Route Retrieval

Each query first undergoes verse-reference parsing (regex) and LLM intent classification extracting entity names and keywords. A signal detector derives six boolean features: (1) book+chapter:verse, (2) book+chapter-only, (3) ≥2 book names, (4) ≥2 person entities, (5) event keyword, (6) place name—via regex and dictionary matching against knowledge-graph entity lists.

A priority-ordered decision tree (Table I) selects among six routes, each combining a subset of four engines—SQL (PostgreSQL), Semantic (Qdrant), Graph (Neo4j traversal), and Cross-Reference (Neo4j CROSS_REFERENCES). R1 resolves exact verse lookups via SQL, bypassing reranking; R2 adds semantic search to chapter-filtered SQL; R3–R6 pair graph traversal for person, event, cross-reference, and place queries with semantic search and SQL. Candidates then undergo deduplication and BGE-Reranker-v2-M3 [11] reranking.

TABLE I

Signal-adaptive Route Overview

Route	Signal Trigger	Engines (weights)
R1	Book + chapter: verse	SQL (1.0)
R2	Book + chapter	SQL (.9) + Semantic (.6)
R3	≥2 persons	Graph (.9) + Semantic (.7) + SQL (.5)
R4	event keyword	Graph (.85) + Semantic (.7) + SQL (.5)

R5	≥2 books	XRef (.85)‖Graph (.75) + SQL (.85) + Semantic (.65)
R6	place name	Graph (.85) + Semantic (.7) + SQL (.5)
FB	(else)	Semantic (1.0)

D. Pluggable LLM Generation

A factory pattern serves multiple providers (Claude, Gemini, OpenAI, Ollama) behind one evidence-first, citation-enforcing prompt. Since the LLM also classifies intent (Fig. 1), swapping models can shift routing on complex queries—a trade-off analyzed in Section IV.

III. EXPERIMENTS

Setup: We evaluate on 100 Traditional Chinese questions (20 per category) with ground truth approved by annotators with biblical expertise: VERSE_LOOKUP, TOPIC, PERSON, EVENT, and GENERAL. This taxonomy tests exact matching, semantic search, entity retrieval, and knowledge integration.

We compare Claude Haiku 4.5 (commercial API) [8] against Gemma 4 E4B (local Ollama) [7] on the same retrieval codebase (temperature=0.1, top_k=5). To isolate the knowledge-graph contribution, we also evaluate a semantic-only baseline bypassing signal- adaptive routing and graph traversal—four configurations in total (Graph/Semantic × Claude/Gemma). In addition to retrieval and RAGAS generation metrics [9], we introduce Answer Coverage. An LLM judge evaluates how well each ground-truth answer point is addressed, assigning scores of 1.0 (covered), 0.5 (partial), or 0.0 (missing), with Coverage representing the mean score. Unlike F1-style correctness, this pure recall metric rewards completeness without penalizing extra detail, effectively complementing faithfulness as a hallucination check.

TABLE II
Overall Performance: Graph RAG vs. Semantic-Only Baseline

Category	Metric	Signal Adaptive		Semantic	
		Claude	Gemma	Claude	Gemma
Retrieval	Hit Rate	0.960	0.950	0.810	0.810
	Recall@5	0.886	0.889	0.758	0.758
	MRR	0.805	0.802	0.647	0.647
Generation	Faithfulness	0.957	0.951	0.899	0.903
	Relevancy	0.815	0.893	0.709	0.774
	Context Recall	0.797	0.791	0.680	0.677
	Coverage	0.750	0.722	0.618	0.628

TABLE III
Answer Coverage by Question Type (Signal-adaptive RAG)

Type	Claude	Gemma	Δ
VERSE	1.000	0.975	−0.025
TOPIC	0.900	0.856	−0.044
PERSON	0.635	0.594	−0.041
EVENT	0.613	0.606	−0.007
GENERAL	0.601	0.580	−0.021

IV. RESULTS AND DISCUSSION

Finding 1: Signal Adaptive Retrieval Drives Quality

Table II shows the signal-adaptive pipeline outperforms a semantic-only baseline on every metric, with decisive retrieval gains: hit rate +15 pp (0.81→0.96), recall@5 +13 pp (0.758→0.886), MRR +16 pp (0.647→0.805). Signal routing and graph traversal successfully retrieve relevant chapters that dense retrieval misses. This improved retrieval outcome propagates to the generation stage, where faithfulness and coverage improve from 0.90 to 0.95 and from 0.60 to 0.74, respectively. The gain is model-independent: Claude and Gemma improve by near-identical margins, isolating retrieval architecture as the dominant quality lever.

Finding 2: Complementary Model Strengths

Holding the pipeline fixed, Claude and Gemma differ by under 4 pp on 6 of 7 metrics; only answer relevancy differs more (7.8pp). Strengths are complementary: Claude leads on faithfulness (0.957 vs. 0.951) and coverage, Gemma on relevancy (0.893 vs. 0.815). Retrieval metrics are identical for deterministic routes (R1, R2) but diverge slightly for R3–R6, where LLM intent classification shapes graph-traversal inputs. Given high-quality context, model differences manifest as generation style, not quality gaps.

Finding 3: Question Type Analysis

Table III shows route-dependent patterns. VERSE scores highest (1.000/0.975) via exact SQL match, with TOPIC close behind (0.900/0.856); PERSON (0.635/0.594), EVENT (0.613/0.606), and GENERAL (0.601/0.580) are hardest—person queries hinge on imperfect entity-mention co-reference, multi-chapter narratives strain fixed-window chunking, and open-ended general questions demand broad knowledge integration. Sparse CROSS_REFERENCES edges (916) further cap the cross-book route. Claude leads on all five types.

Finding 4: Cost-Quality Tradeoff

Gemma 4 E4B runs locally via Ollama at zero API cost with full data privacy—critical for sensitive corpora. Since neither model dominates, the local option is viable: the key investment shifts from model selection to retrieval infrastructure, where the signal-adaptive router tailors an engine combination to each query type.

V. CONCLUSION

We designed, deployed, and evaluated a signal-adaptive RAG architecture for Traditional Chinese Bible QA. Our evaluation suggests that this approach offers a potentially viable and cost-effective RAG methodology for specialized religious domains. While further validation on larger, more diverse datasets is required to fully verify these preliminary effects, our architecture indicates that signal-adaptive routing may offer a means to decouple domain-specific performance from a reliance on expensive, large-scale commercial APIs. Furthermore, the core mechanics of this routing framework may hold broader utility for industrial applications facing similar data-siloing and cost constraints. Future work will explore and benchmark this approach against agentic, LLM-driven routing mechanisms.

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